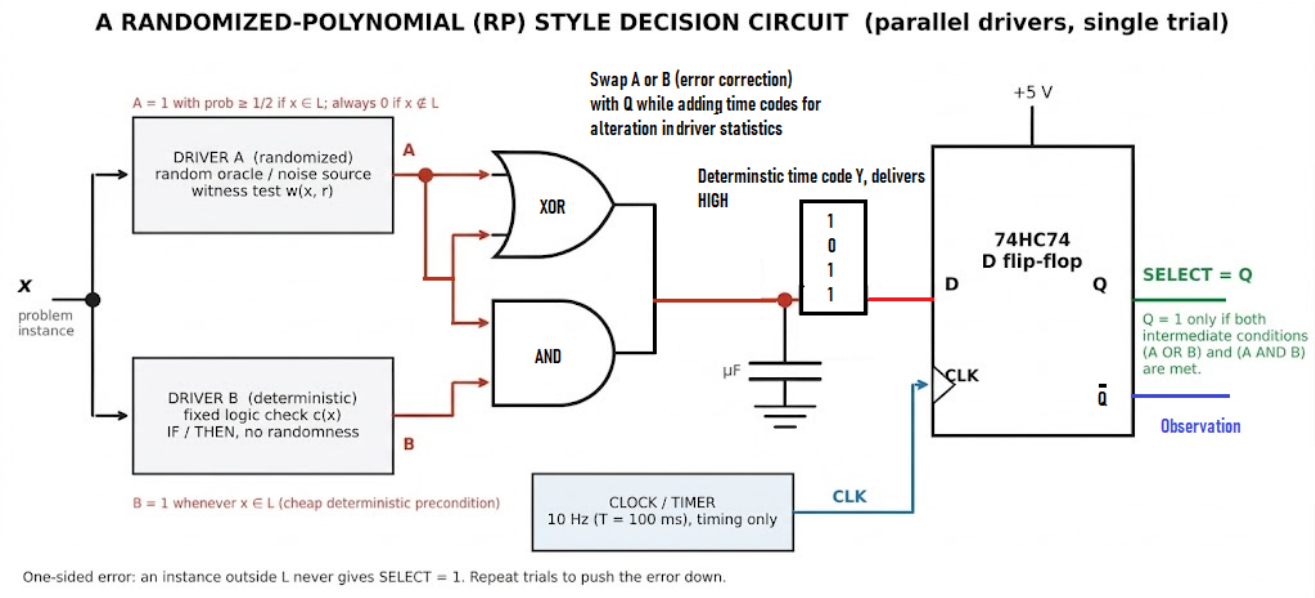
The Protocol: Energy-Level Switching, Measurement and Closed Timelike Curves

How one trial of the RP-style circuit works as a physical protocol, and what its feedback loop has to do with time loops



The circuit this protocol is built on

1. The protocol in five steps

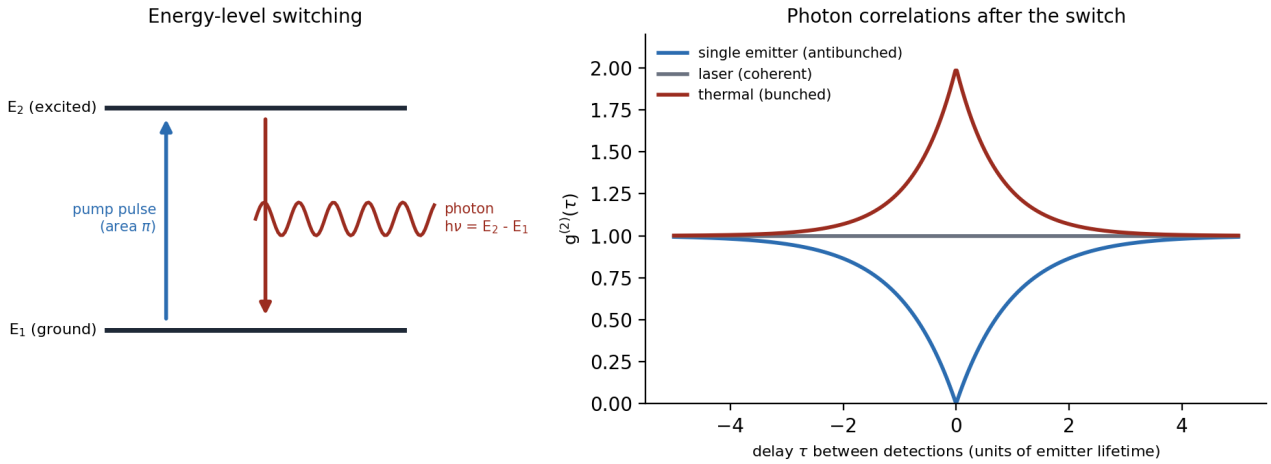
Step	Circuit element	Physical event	Result
1. Prepare	Input x, clock, time code Y = 1011	Choose the excitation and the allowed time bins (pump in bins 1, 3, 4; skip bin 2).	Defines when and how strongly electrons are driven.
2. Switch levels	Drivers A and B	An electron is driven from E_1 to E_2. On relaxation it emits a photon with $h\nu = E_2 - E_1$.	Photon emission, random (A) or metered (B).
3. Gate	AND / XOR gating	Emission or counting is allowed only when the required conditions coincide.	Combined condition, read here as A AND B.
4. Measure	RC node, 74HC74, CLK	A detector turns photons into a click. On the clock edge the flip-flop latches it.	Q = SELECT = 1 (accept) or 0.
5. Repeat	Clock windows	Run k independent trials; accept if any trial gives Q = 1.	Worst-case error $(1/2)^k$.

2. Photons controlled by electron energy-level switching

A photon is emitted when an electron drops between two energy levels, and its energy is fixed by the gap. What the circuit's two drivers control is *when* the switch happens, and that sets the photon statistics.

- **Driver A, random excitation.** Thermal or noisy pumping makes each emitter switch at a random time. Independent emitters give Poissonian counts (Fano factor F = 1). Thermal fluctuations on top push F above 1.
- **Driver B, deterministic excitation.** A resonant pulse with area π inverts a two-level system with probability close to 1, since the excited-state probability is $\sin^2(\theta/2)$. After emitting, the emitter is in its ground state and cannot emit again until it is re-excited, so photons tend to arrive one at a time. This is antibunching, with $g^{(2)}(0)$ approaching 0 and F < 1.

- **Clock and time code.** They decide which bins are pumped, so they fix the counting window and the temporal pattern of the emission.



Left: the pump moves an electron from E_1 to E_2, and relaxation emits a photon. Right: the correlation function $g^{(2)}(\tau)$ for three kinds of source.

3. What measurement does

Photon-number outcomes follow the Born rule, $P(n) = |\langle n|\psi\rangle|^2$. Right after emission the emitter and the photon can be entangled, so detecting the photon also updates what you know about the emitter. The measurement chain adds its own effects:

- A detector with efficiency η thins the stream, so $F_{ph} = 1 - \eta(1 - F_e)$. Regularity from Driver B survives only as far as η allows.
- The click is an irreversible, thresholded event. That is why a clocked flip-flop is a fair model: it samples at the clock edge and holds the result.
- Dark counts (clicks with no photon) are false accepts. The ideal RP circuit has none, so dark counts are where the analogy breaks.

4. The feedback loop and closed timelike curves

The diagram's note, “swap A or B with Q while adding time codes”, describes feeding the output Q back into the drivers. That loop is what connects the circuit to the idea of a **closed timelike curve (CTC)**: a worldline that returns to its own past. CTCs appear as solutions of general relativity (for example Gödel's rotating universe and some wormhole geometries). There is no evidence that they exist, and Hawking's chronology protection conjecture argues that physics forbids them.

Deutsch's model. In David Deutsch's 1991 quantum model, a system going around a CTC must come back in the same state it left in. The state ρ of the looping system satisfies a fixed-point condition:

$$\rho_{CTC} = \mathrm{Tr}_{other}\left[U\left(\rho_{other} \otimes \rho_{CTC} \right) U^{\dagger} \right]$$

Grandfather paradox as a bit. If the loop must flip a bit (x = NOT x), no definite value is consistent. Deutsch's rule picks the 50/50 mixture, which is the only self-consistent state. It echoes the 1/2 success threshold of the randomized driver, but this is an analogy and not a derivation.

Complexity. Aaronson and Watrous (2009) showed that with CTCs, both classical and quantum computers would solve exactly the problems in PSPACE, a class that contains RP. **Experiments.** Ringbauer and colleagues (2014) simulated CTC behaviour with photons using post-selection. This reproduces the mathematics, not an actual loop through time.

In the circuit, the feedback from Q to the pump is an ordinary electrical loop, not a time loop. It resembles the CTC fixed-point condition only in that the output must be consistent with the input it helps produce.

5. Parameters of the system

Parameter	Role	Where it appears
$E_2 - E_1$	Photon energy and wavelength	Level switching

Parameter	Role	Where it appears
τ (emitter lifetime)	Emission rate; recovery time between photons	$g^{(2)}(\tau)$ dip width
Pulse area θ	How deterministic the excitation is (π gives inversion)	Driver B
F_e, η	Electron noise and overall efficiency, giving F_{ph}	Sections 2 and 3
Dark-count rate	Sets the false-accept rate	Detector and flip-flop
Clock period T	Counting window; 100 ms here, ns or shorter in real devices	CLK
Time code Y	Which bins are pumped	Pump timing
k (trials)	Number of repetitions; error $(1/2)^k$	Repeat step
Loop consistency	Fixed-point condition on the fed-back state	CTC reading of the feedback

6. What is established and what is speculative

Status	Content
Established physics	Level-switching emission; Poissonian, thermal and antibunched statistics; the Born rule; the thinning relation for detector efficiency; one-sided error and $(1/2)^k$ repetition.
Model-dependent	Deutsch's fixed-point rule for CTCs. Other models (post-selected CTCs, for example) give different predictions, and Deutsch's model has been criticised for its nonlinear features.
Speculative	Whether CTCs exist at all, and any link between the circuit's feedback and a real time loop.

7. References to check

- D. Deutsch, “Quantum mechanics near closed timelike lines”, Physical Review D, 1991.
- S. Aaronson and J. Watrous, “Closed timelike curves make quantum and classical computing equivalent”, Proceedings of the Royal Society A, 2009.
- M. Ringbauer et al., “Experimental simulation of closed timelike curves”, Nature Communications, 2014.
- S. Hawking, “Chronology protection conjecture”, Physical Review D, 1992.

These citations are from memory and have not been checked against the publications.